

No books, notes, computational devices, etc. are allowed. Use clear handwriting and justify your answers carefully. Please write your anonymous identification number on each sheet of paper.

1. Let M, M' be right R -modules and N, N' be left R -modules, where R is a ring with unity. Suppose $f : M \rightarrow M'$ and $g : N \rightarrow N'$ are R -homomorphisms. If f and g are surjective, show that $f \otimes g$ is also surjective. Does the same hold replacing surjective with injective?
2. If G is a group with $|G| = 2^r$ for some positive integer r , show that there exists a series of normal subgroups

$$1 = G_0 \leq \cdots \leq G_r = G$$

such that $|G_j| = 2^j$ for $0 \leq j \leq r$.

3. Obtain the Smith normal form and rank for the following matrix over the principal ideal domain $\mathbb{Q}[x]$:

$$\begin{pmatrix} -x-3 & 2 & 0 \\ 1 & -x & 1 \\ 1 & -3 & -x-2 \end{pmatrix}$$

4. Let M be an Artinian left R -module. Prove that every quotient module of M is finitely cogenerated.
5. Let G be a group of order 108. Show that there exists a normal subgroup of order 27 or 9.
6. Suppose R is a principal ideal domain. Let M be an R -module and let N be a given submodule of M . The submodule N is called *isolated* in M if N is equal to

$$\{x \in M \mid \exists r \in R \setminus \{0\} \text{ such that } rx \in N\}.$$

Let M be a free R -module. Suppose that N is an isolated submodule of M and suppose that M/N is finitely generated. Show that M/N is a free R -module, and show that there exists a free submodule L of M such that $M = N \oplus L$.